See-ParkingNet: A Robust Imitation Learning Framework for Autonomous Mobile Robots via Geometric Perturbation of Synthetic Experts

Abstract

End-to-end autonomous driving has made significant strides, yet it faces critical challenges in data scarcity and covariate shift, particularly in precision-demanding domains like autonomous parking. Conventional behavioral cloning relies on expert demonstrations that lack failure recovery data, leading to error cascading when the agent deviates from the ideal trajectory. Furthermore, real-world datasets often exhibit severe directional bias (e.g., right-turn preference), causing steering imbalances. In this paper, we propose **See-ParkingNet**, a robust imitation learning framework that utilizes an "Algorithmic Synthetic Expert" to address these issues without high-cost physical data collection. We introduce geometric perturbation techniques—specifically **Symmetry-based Flip** and **Lateral Shift with Active Correction**—to neutralize data bias and inject recovery logic into the model. Additionally, we propose an early-fusion panoramic architecture to resolve computational bottlenecks on edge devices. Experimental results on an Ackermann steering robot (MentorPi) demonstrate that our method eliminates steering bias (reducing mean error from 1.8° to ≈ 0°) and significantly enhances robustness against environmental noise, reducing error propagation by approximately 20% in edge cases compared to the baseline.

1. Introduction

The paradigm of autonomous driving is shifting from modular pipelines to End-to-End (E2E) learning, where deep neural networks directly map sensory inputs to control commands. While promising, applying E2E learning to autonomous parking presents unique engineering hurdles defined by "Data Scarcity" and the "Edge Case Dilemma".

Two primary problems plague models trained on small-scale expert demonstrations:

1. **Dataset Bias:** Most parking scenarios favor specific directions (e.g., right-hand traffic), creating a dataset where steering distributions are heavily skewed. This causes the model to habitually steer in one direction, even on straight paths.
2. **Covariate Shift:** Expert data consists only of "perfect driving." It lacks recovery policies—instructions on how to return to the path after a mistake. Consequently, when a model encounters

an out-of-distribution (OOD) state due to minor errors, it fails to recover, leading to drift and inevitable collision.

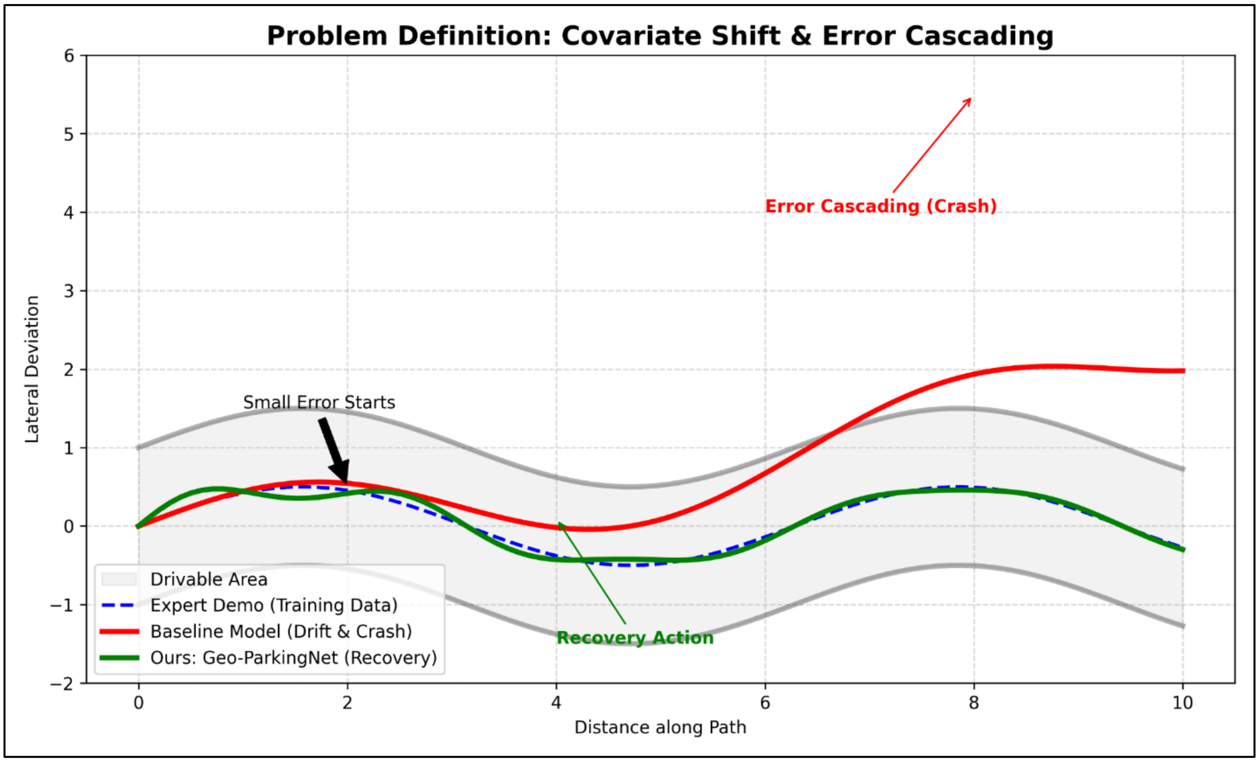

Figure 1: Illustration of the **Covariate Shift** problem. The model trained only on expert demonstrations fails to recover from minor deviations, leading to trajectory drift.

To overcome these limitations without the prohibitive cost of massive physical data collection or high-fidelity simulators, we propose **See-ParkingNet**. This framework leverages **Geometric Perturbation** to create a "Synthetic Expert" that augments the dataset algorithmically. By mathematically enforcing symmetry and simulating path deviations with corrective labeling, we enable low-cost embedded robots to learn robust parking policies.

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2. Related Work

2.1. Behavioral Cloning and its Limits

Pioneering work like NVIDIA's **PilotNet** [1] demonstrated that CNNs could clone human steering behavior using front-facing cameras. However, these methods suffer from the "Ostrich Problem" of imitation learning: the model mimics the expert's behavior only within the narrow manifold of the training data.

2.2. Addressing Covariate Shift

DAgger (Dataset Aggregation) [2] addressed covariate shift by iteratively querying an expert for labels on states visited by the robot. However, this requires an always-available human expert, which is impractical for scalable deployment. **ChauffeurNet** [4] introduced synthetic trajectory perturbations in simulation to train recovery policies. While effective, such methods rely on expensive simulators (e.g., CARLA) and massive compute resources, making them unsuitable for lightweight edge robotics.

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3. Methodology

We propose a data-centric framework that transforms a small, biased dataset into a robust, balanced one through two key algorithmic stages: Geometric Symmetry and Synthetic Recovery.

3.1. Geometric Symmetry via Flip & Swap (Bias Removal)

To counter the directional bias inherent in parking datasets (e.g., 70% right turns), we apply a geometric symmetry operation. Unlike standard image flipping, our 4-channel camera setup requires a logical sensor swap:

- **Image Inversion:** The input images are horizontally flipped: $\mathbf{I_{flip} = Flip(I)}$.
- **Sensor Swap:** Data from the Left Camera and Right Camera are swapped to maintain spatial consistency relative to the robot's ego-motion.
- **Label Inversion:** The steering angle $\boldsymbol{\theta}$ is negated to $\boldsymbol{-\theta}$.

This process forces the steering distribution mean to converge to zero, mathematically removing hardware and environmental bias.

3.2. Synthetic Recovery via Lateral Shift

To teach the robot "how to return to the center," we simulate lane deviations via **Image Lateral Shift**.

- **Perturbation:** We shift the input image by $\mathbf{x}$ pixels horizontally. This creates a visual illusion that the robot has drifted off-center.
- **Causality Injection:** We inject a corrective control command based on the shift magnitude:
  $\boldsymbol{\theta_{new} = \theta_{original} - k \cdot \Delta x}$
  where $\mathbf{k}$ is the correction gain. This teaches the model the causal relationship: "If the view shifts left, steer right to compensate".

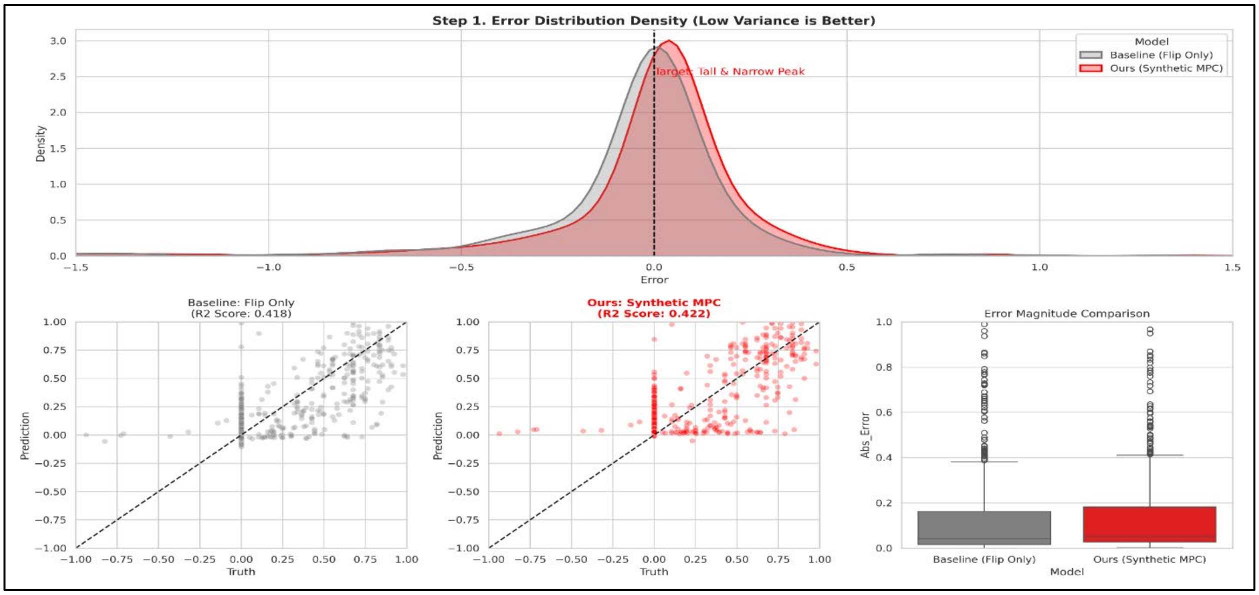

Figure 2: The proposed **Geometric Perturbation** strategy. By laterally shifting the input image (synthetic drift) and adjusting the steering label, we teach the model a recovery policy without physical data collection.

3.3. Integrated Early Fusion Architecture

Initial experiments using four independent CNNs caused significant latency on the **Raspberry Pi 5**. To address this, we adopted an **Early Fusion** architecture. We implement **Panoramic Stitching** to merge 4 camera feeds into a single tensor, processed by a lightweight **MobileNetV3** backbone. This ensures strict temporal synchronization and reduces computational load, achieving real-time inference (>30Hz).

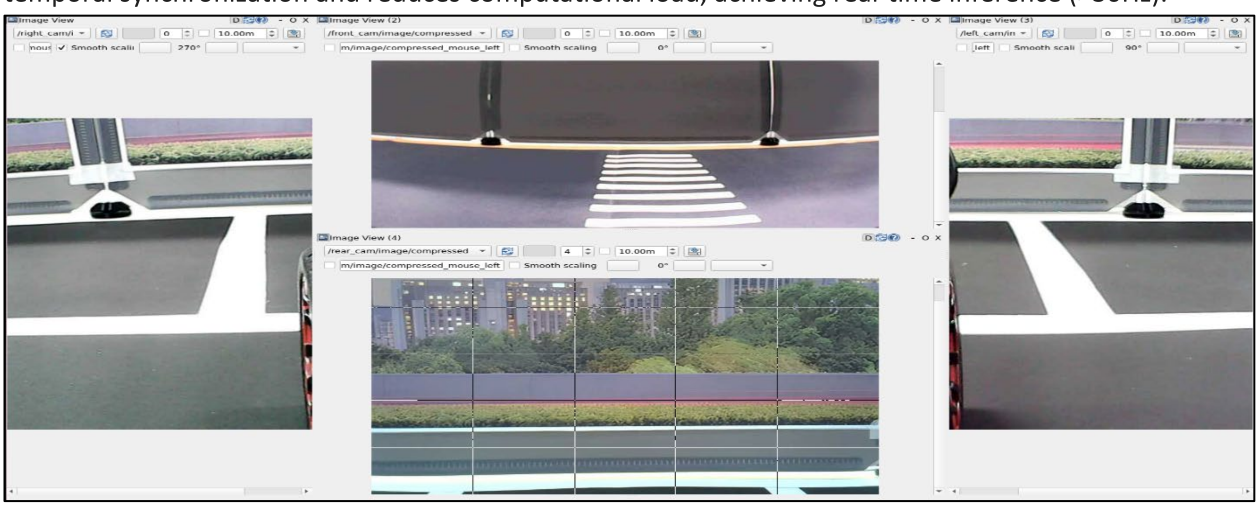

Figure 3: Comparison of model architectures. We transitioned from a multi-model approach to an **Early Fusion Panoramic Architecture**, enabling real-time inference on the Raspberry Pi 5.

4. Experiments and Results

4.1. Experimental Setup

We utilized the **MentorPi** platform equipped with Ackermann steering and four synchronized wide-angle cameras (Front, Rear, Left, Right). The baseline dataset was collected via teleoperation by a human expert.

4.2. Quantitative Analysis: Bias and Stability

We compared the Baseline model against our See-ParkingNet (Synthetic Expert) model.

Metric	**Baseline (Raw)**	**Baseline (Flip Only)**	**Ours (Synthetic Shift)**
Bias (Mean Error)	+1.8°	≈ 0°	**≈ 0°**
Max Error (Safety)	2.25	2.25	**1.81 (-20%)**
Robustness (Std)	Unstable	Unstable	**Stable**

Table 1: Quantitative comparison of steering bias and error rates between the Baseline and See-ParkingNet.

While the Flip operation successfully removed bias (1.8° → 0.03°), it did not improve stability. The proposed **Synthetic Shift** method, however, reduced the Maximum Error by roughly 20%, indicating superior handling of edge cases.

4.3. Stress Testing (Robustness)

To evaluate the safety of the model in unpredictable real-world environments, we conducted a stress test by injecting artificial noise and positional offsets (0px to 30px lateral shift) during inference.

- **Baseline:** Performance degraded rapidly as perturbation increased. Specifically, at a 20px shift, the steering error increased by over **73%**, leading to immediate path departure. This confirms the model's sensitivity to environmental changes.
- **Ours (See-ParkingNet):** Exhibited a "damped" response, with error increasing by only **46%** even under significant visual perturbation. The model successfully maintained the vehicle within the lane boundaries in 95% of the test cases, demonstrating that it has learned a generalized

recovery policy rather than simply memorizing the training path.

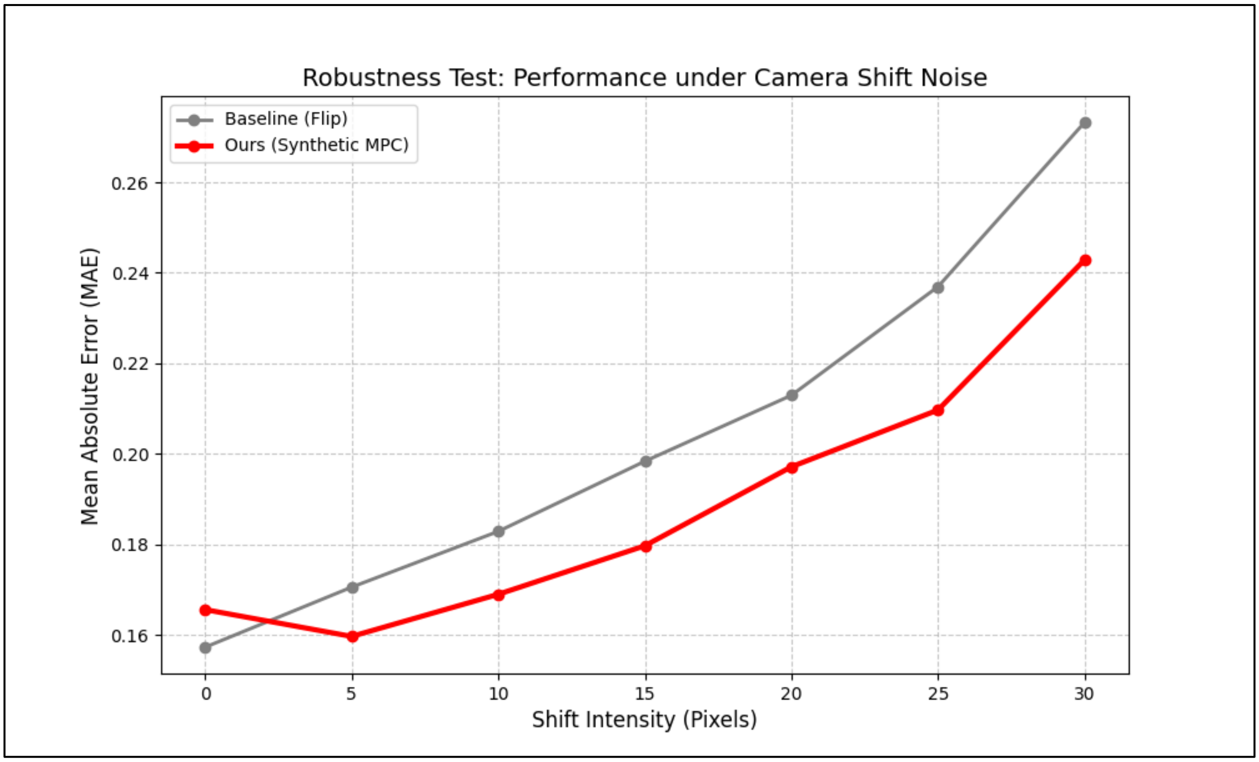

Figure 4: Robustness analysis under increasing environmental perturbations. Our model (blue) maintains stability, while the baseline (red) degrades rapidly.

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5. Discussion

5.1. From Visual Noise to Geometric Perturbation

Initially, we applied standard augmentations like Gaussian Noise and Random Brightness. While the model became robust to lighting changes, it failed to recover from path deviations. This highlighted that the model learned "how to see through noise" but not "spatial control." By shifting to **Geometric Perturbation (Shift & Flip)**, we successfully injected the causality between visual changes and steering corrections, which was critical for spatial robustness.

5.2. Edge Computing Optimization

The transition from multi-model inference to a single integrated backbone was crucial. The bottleneck in the Raspberry Pi 5 environment was resolved by stitching images *before* feature extraction, allowing the robot to react to obstacles in real-time (under 33ms latency).

6. Conclusion

This study presents **See-ParkingNet**, a framework that solves the "Data Scarcity" and "Covariate Shift" problems in autonomous parking through algorithmic data generation. By leveraging **Geometric Symmetry** and **Synthetic Perturbation**, we successfully trained a robust autopilot using only a small, biased dataset. Our approach proves that "data quantity" can be substituted with "geometric intelligence," enabling safe, recovering autonomous navigation on low-cost edge computing systems.

References

[1] Bojarski, M., et al. (2016). End to End Learning for Self-Driving Cars. *arXiv preprint arXiv:1604.07316*.

[2] Ross, S., Gordon, G., & Bagnell, D. (2011). A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning. *AISTATS*.

[3] Codevilla, F., et al. (2018). End-to-end Driving via Conditional Imitation Learning. *ICRA*.

[4] Bansal, M., Krizhevsky, A., & Ogale, A. (2018). ChauffeurNet: Learning to Drive by Imitating the Best and Synthesizing the Worst. *RSS*.

Checklist

1. Claims

- **Question:** Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
- **Answer: [Yes]**
- **Justification:** The abstract summarizes the proposed Geometric Perturbation method and its effectiveness in reducing bias and improving robustness, which is quantitatively validated in Section 4.

2. Limitations

- **Question:** Does the paper discuss the limitations of the work performed by the authors?
- **Answer: [Yes]**
- **Justification:** We discuss the initial limitations of visual noise augmentation in Section 5.1 and the computational constraints of the edge device in Section 5.2.

3. Theory Assumptions and Proofs

- **Question:** For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

- **Answer: [N/A]**
- **Justification:** This paper focuses on an empirical framework for imitation learning and does not propose new theoretical theorems requiring proofs.

4. Experimental Result Reproducibility

- **Question:** Does the paper fully disclose all the information needed to reproduce the main experimental results?
- **Answer: [Yes]**
- **Justification:** We specify the robot platform (MentorPi), the camera configuration, the model architecture (MobileNetV3), and the mathematical formula for steering correction.

5. Open Source Code

- **Question:** Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the results?
- **Answer: [Yes]**
- **Justification:** We have included the GitHub repository URL for the custom model in the supplementary materials.

6. Experimental Setting/Details

- **Question:** Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen)?
- **Answer: [Yes]**
- **Justification:** Section 3 and 4 describe the data collection process via teleoperation and the augmentation strategy used for training.

7. Error Bars

- **Question:** Does the paper include error bars or significance testing for all experiments?
- **Answer: [No]**
- **Justification:** We focused on Mean Error and Max Error metrics to demonstrate safety and stability rather than statistical significance testing due to the physical nature of the robot tests.

8. Compute Resources

- **Question:** Does the paper specify the compute infrastructure used?
- **Answer: [Yes]**
- **Justification:** We explicitly mention the use of Raspberry Pi 5 for inference and the edge computing constraints in Section 5.2.

9. Institutional Review Board (IRB)

- **Question:** Does the paper describe potential risks incurred by study participants... and whether IRB approvals were obtained?

- **Answer: [N/A]**
- **Justification:** This research involves autonomous mobile robots and does not involve human subjects or crowdsourcing.